ATHENA TECH

DATA SCIENCE RESOURCES

E-commerce Projects

Project 1. Customer Lifetime Value (CLV) Prediction

Problem Statement: Customer acquisition is expensive. To maximize ROI, businesses must identify customers with high long-term value and target them with retention-focused strategies.

Objective: Build a regression or probabilistic model to predict Customer Lifetime Value (CLV) using transactional, behavioral, and demographic data.

Dataset Source

- Online Retail Dataset (UCI / Kaggle)
- Simulated e-commerce CRM logs

Techniques & Concepts

- RFM analysis (Recency, Frequency, Monetary)
- Linear Regression, XGBoost, or Gamma-Gamma model
- LTV segmentation for marketing campaigns
- CLTV scoring and visualization

Tools & Libraries: Python, Lifetimes (for probabilistic modeling), Pandas, Scikit-learn, Matplotlib, Plotly

Expected Outcome: A model that segments customers by predicted LTV (High, Medium, Low) to support retention, reactivation, and upselling strategies.

Project 2. Shopping Cart Abandonment Predictor

Problem Statement: A large percentage of users add items to their cart but never complete the purchase. Identifying the behavioral patterns behind this can significantly boost conversions.

Objective: Build a binary classification model that predicts if a customer will abandon their cart before purchase, based on session behavior, time spent, clicks, and items viewed.

Dataset Source

- E-commerce Behavior Data
- Session log exports from Google Analytics or Mixpanel

Techniques & Concepts

- Binary classification (Logistic Regression, Random Forest)
- Session-based features and clickstream analysis
- Class imbalance handling (SMOTE, threshold tuning)
- Behavioral funnel visualization

Tools & Libraries: Python, Pandas, Scikit-learn, XGBoost, Matplotlib, Streamlit (for demo app)

Expected Outcome: A real-time predictor for abandonment that can trigger retargeting actions like email reminders, discount offers, or chatbot prompts.